Rizwan Ali Shah

Ph.D. in Electronics Engineering | Computer Vision Researcher

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About Me

Experienced researcher specializing in Computer Vision with a deep focus on Deep Learning, Machine Learning, and Self-Supervised Learning (SSL). Proven expertise in developing unsupervised and self-supervised architectures for complex industrial applications, including multi-class anomaly detection and segmentation using synthetic data augmentation. Strong background in handling massive multimodal datasets across large-scale R&D frameworks.

Education

2021–2025 **Ph.D. in Electronics Engineering**, *Chungbuk National University*, Cheongju, South Korea, *Specialization: Computer Vision, Deep Learning, Image Processing*

Thesis **Self-Supervised Deep Learning Technique for Multi-Class Industrial Anomaly Detection Based on Synthetic Anomaly Insertion** (Advisor: Prof. HyungWon Kim)

2015–2017 **Master of Philosophy (M.Phil.) in Electronics**, *Quaid-i-Azam University*, Islamabad, Pakistan, *Specialization: Digital Image Processing, Pattern Recognition*

Publications & Research Manuscripts

Peer-Reviewed Journal Publications

2025 **Rizwan Ali Shah**, et al. "Self-Supervised Image Anomaly Detection Through Diverse Pseudo Anomaly Insertion." *IEEE Access* (first author).

2025 **Rizwan Ali Shah**, et al. "Patch-Shuffled Superpixel Synthetic Anomaly Insertion in Autoencoder Foreground Extracted Regions for Improved Visual Anomaly Detection." *IEEE Access* (co-author).

2023 **Rizwan Ali Shah**, et al. "Two-stage coarse-to-fine image anomaly segmentation and detection model." *Image and Vision Computing* (first author, Elsevier, JCR IF Top 20%).

Manuscripts Under Review (Co-Author)

2026 "RAUDI: A Dual-Head Framework Unifying Unsupervised and Supervised Reconstruction for Industrial Anomaly Detection." *Image and Vision Computing*.

2026 "CSR: Continual Subspace Refinement for One-Shot Industrial Anomaly Detection." *IEEE Transactions on Industrial Informatics*.

Peer-Reviewed International Conference Papers

2021 **Rizwan Ali Shah**, et al. "Improving Performance of CNN Based Vehicle Detection and Tracking by Median Algorithm." *2021 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia).*

2022 **Rizwan Ali Shah**, , et al. "Enhancement of Anomaly Detection Using Two-Stage Anomaly Segmentation Model." *The 8th International Conference on Next Generation Computing 2022*.

2023 **Rizwan Ali Shah**, et al. "Advanced Color Pseudo Anomaly to Enhance Learning of Convolutional Neural Network Models." *2023 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia).*

2024 **Rizwan Ali Shah**, et al. "Patch-Oriented Anomaly Detection and Segmentation Model." *Korean Institute of Communication and Communications Artificial Intelligence Conference, 2024*.

2025 **Rizwan Ali Shah**, et al. "Multi-Class Segmentation Network for Robust Class-Specific Anomaly Detection." *Summer Conference of the Korea Electronic Engineering Association, 2025*.

Working Papers & Preparing for Submission

Working — "A Multi-Signal Dual-Decoder Framework for Unsupervised Industrial Anomaly Detection." (*Co-Author, Manuscript in Preparation*).

Working — "Federated Learning With Orthogonally Enhanced Aligned Client Self-Training." (*Co-Author, Manuscript in Preparation*).

Working — **Rizwan Ali Shah**, et al. "Multi-Channel Agnostic Self-Supervised Multi-Class Anomaly Detection for Industrial Datasets." (*First Author*).

Working — **Rizwan Ali Shah**, et al. "Exploiting Geometric Symmetries to Enhance Structural Anomaly Segmentation in Industrial Vision." (*First Author*).

Research & Professional Experience

2021–Present — **Research Assistant / AI Research Engineer**, *Chungbuk National University Industry-Academic Cooperation Foundation*, Cheongju, South Korea

Developed and evaluated scalable vision architectures for critical government-funded initiatives:

- **Ministry of SMEs and Startups Project (66M KRW):** Designed a self-learning object detection camera system for autonomous driving, pioneering the pseudo-anomaly insertion pipelines that minimized edge-case false positives.
- **IITP National Project (1.3B+ KRW):** Pioneered Mobile Self-Learning Recursive Neural Network Processor Technologies; engineered localized, patch-level resource-efficient architectures for constrained hardware environments.
- **NRF Phase 4 BK21 Project (2.6B+ KRW):** Investigated advanced deep representation models, optimizing multi-scale spatial transformers and contrastive self-supervised pipelines for zero-shot generalization.

2018–2021 — **Lecturer in Electronics Department**, *Quaid-i-Azam Univeristy*, Islamabad, Pakistan

Delivered foundational and advanced undergraduate courses, bridging theoretical frameworks with lab instruction:

- Formulated and taught core course curricula across Computer Science, Applied Electronics, and Image Processing.
- Supervised undergraduate laboratory design projects, emphasizing computational logic modeling, Python setups, and algorithmic problem-solving paradigms.

Technical Skills

Deep Learning — Self-Supervised Learning (SSL), Foundation Models, Representation Learning, Unsupervised Anomaly Detection, Generative Data Augmentation, Vision Transformers (ViT), CNNs.

Skills — Python, PyTorch, TensorFlow, OpenCV, MATLAB, LaTeX, Git, Linux

Honors & Academic Awards

2024 — **MSISLAB Scholarship & Corporate Job Offer** – Awarded for industrial-academic technology excellence and model optimization innovation.

2022 — **Best Poster Award** – The 8th International Conference on Next Generation Computing 2022 for the paper on two-stage structural anomaly segmentation.

2021–2025 — **Brain Korea 21 (BK21) Graduate Fellowship** – Awarded by the National Research Foundation (NRF) of Korea for outstanding doctoral research track contribution.

Academic Supporters / References

Primary Reference — **Dr. HyungWon Kim** | Professor, Electronics Engineering, Chungbuk National University, South Korea. Email: `hwkim@cbnu.ac.kr`

Reference 2 — **Dr. M. Naeem Ali Bhatti** | Associate Professor, Electronics, Quaid-i-Azam University, Pakistan. Email: `nbhatti@qau.edu.pk`